Quality Control & Management

Introduction

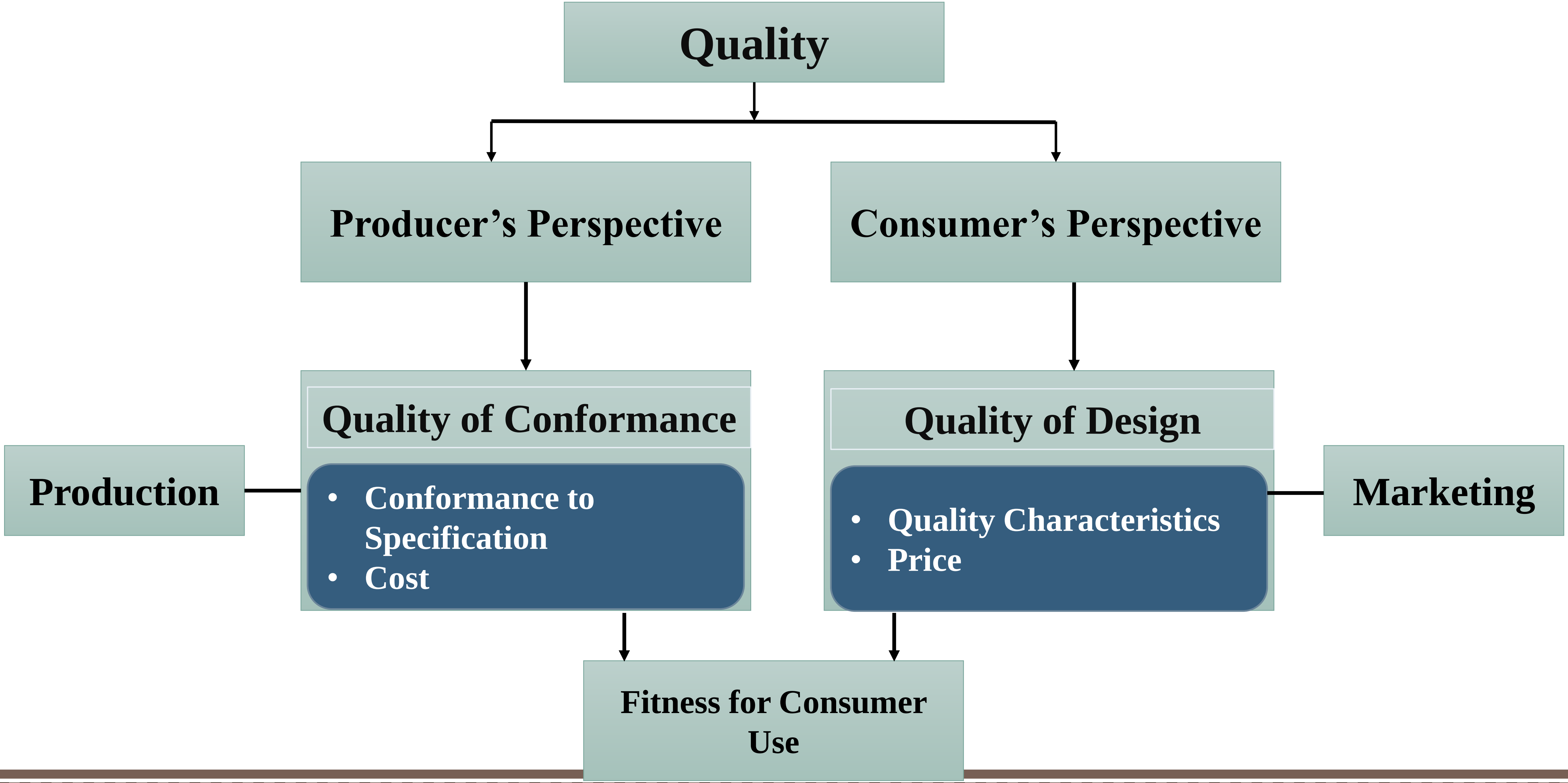
Definition

Quality refers to the characteristic of a product or service that defines its ability to consistently meet or exceed customer expectations.

Q = P / E

Q = Quality
P = Performance
E = Expectation

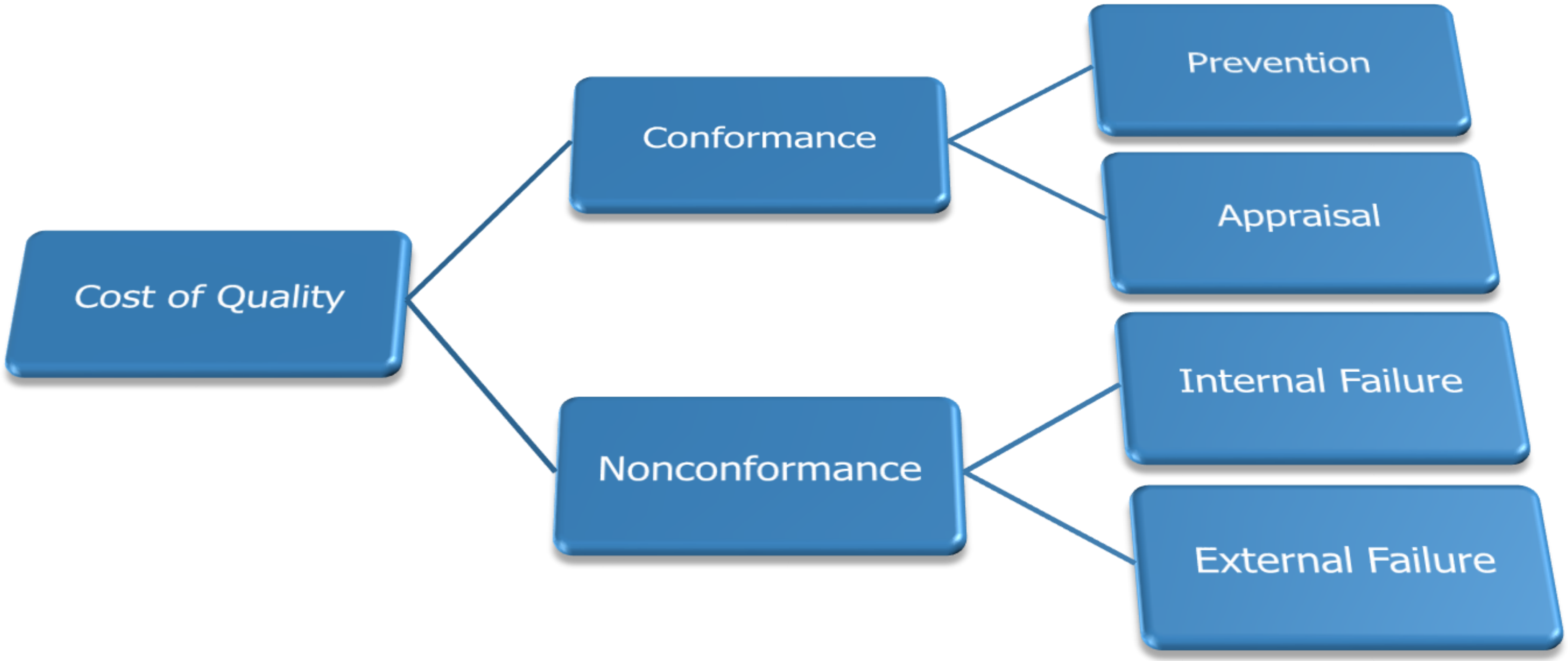
"Fitness for use". Fitness is defined by the customer. Which typically means its Performance, Conformance, Durability, Safety, Reliability.


Quality
Producer's Perspective
Consumer's Perspective
Quality of Conformance
Conformance to Specification
Cost
Quality of Design
Quality Characteristics
Price
Production
Marketing
Fitness for Consumer Use

Consequence of Poor Quality

- Lower productivity, rework, rejection
- Loss of productive time
- Loss of material
- Loss of business organization image
- Liability warranty, replacement, repair

Cost of Quality



Classification of Cost of Quality

Basic Tools of Quality

Seven Basic Tools of Quality is a designation given to a fixed set of graphical techniques identified as being most helpful in troubleshooting issues related to quality.

They are called basic because they are suitable for people with little formal training in statistics and because they can be used to solve the vast majority of quality-related issues.

Seven Basic Tools of Quality

- Check Sheet
- Stratification Analysis
- Control Chart
- Pareto Chart
- Scatter Diagram
- Cause and Effect Diagram
- Histogram

Check Sheet

A generic Tool which can be used for collection and analysis of data – A structured and prepared form that can be adapted for wide variety of issues

When to Use a Check Sheet

- **When data can be observed and collected repeatedly by the same person or at the same location**
- **When collecting data on the frequency or patterns of events, problems, defects, defect location, defect causes**
- **When collecting data from a production process**

Example of Check Sheet

	Complaint Checksheet					
	Mon	Tues	Wed	Thurs	Fri	Total
No. Users	123	110	130	135	128	626
Complaints						
No. Complaints	17	20	24	21	23	105
Percent of users complaining	14%	18%	18%	16%	18%	17%

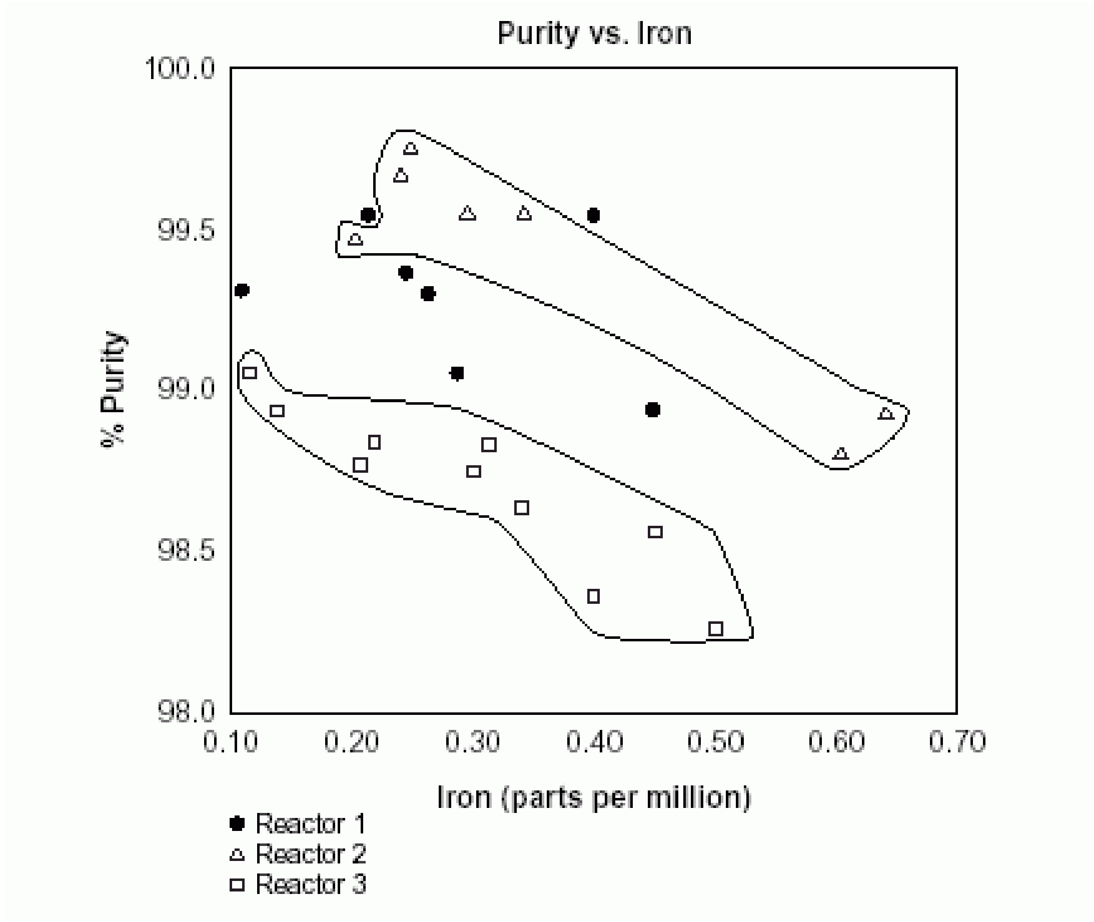
Stratification Analysis

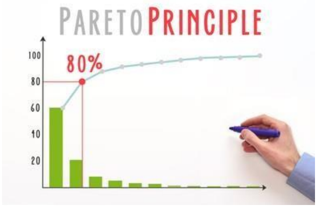
Stratification is a technique used in combination with other analysis tools. When data from variety if source have been lumped together, meaning of the data can be impossible to see. This techniques separate the data

When to Use Stratification Analysis

- Before collecting data
- When data come from different source or condition, such as shifts, days of the week, supplier groups
- When data analysis may require separating different sources or condition

Example of Stratification





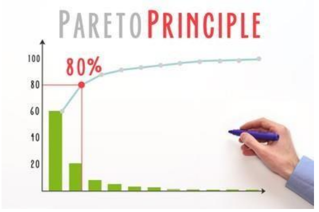
Pareto Chart

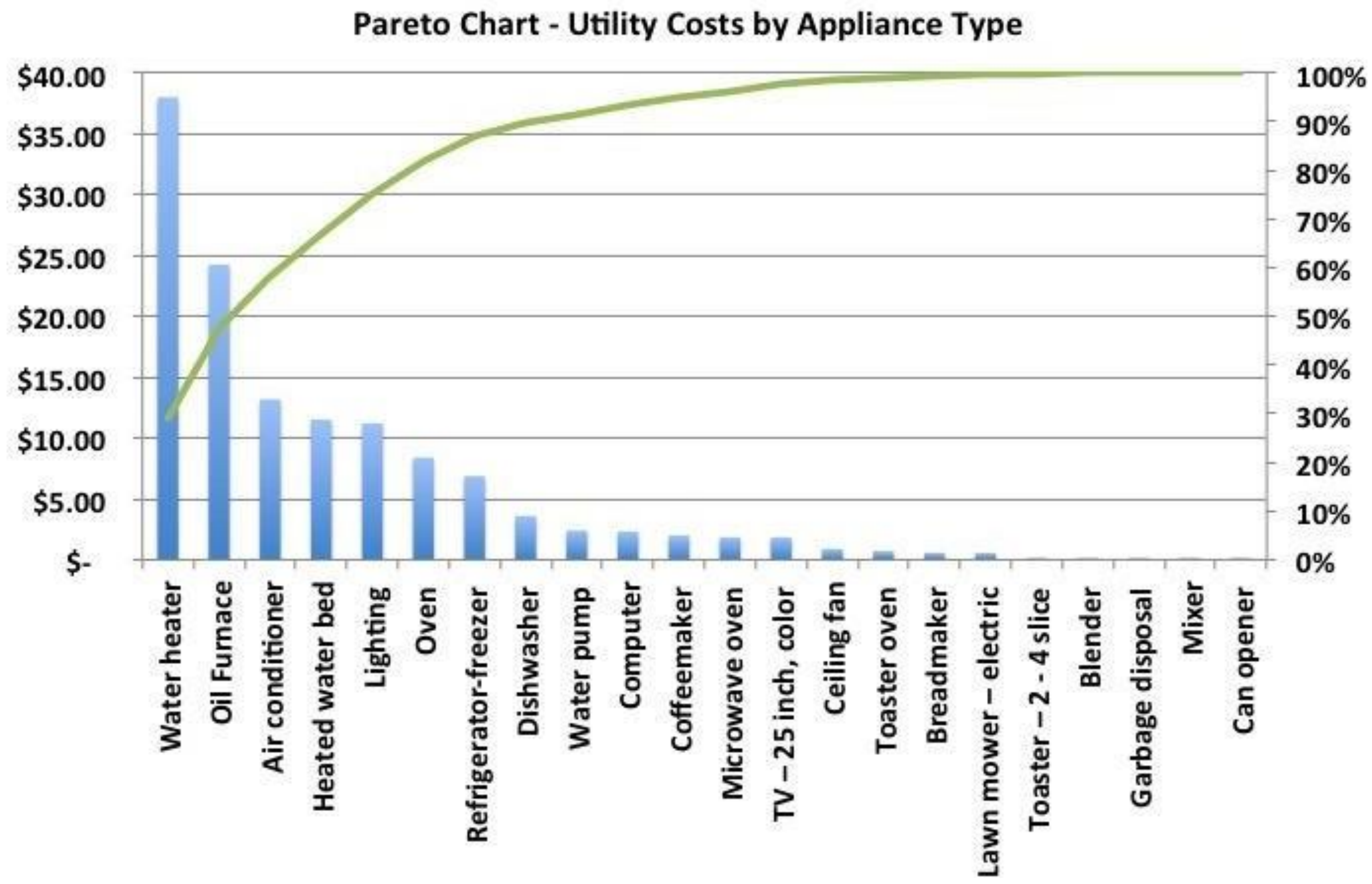
A graphical technique used to identify the significance of individual factors

When to Use a Pareto chart

- **When analyzing data about the frequency of problems or causes in a process**
- **When there are many problems or causes and you want to focus on the most significant**
- **When analyzing broad causes by looking at their specific components**
- **When communicating with others about your data**

Example of Pareto Chart



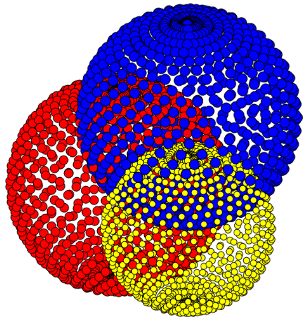


Scatter Diagram

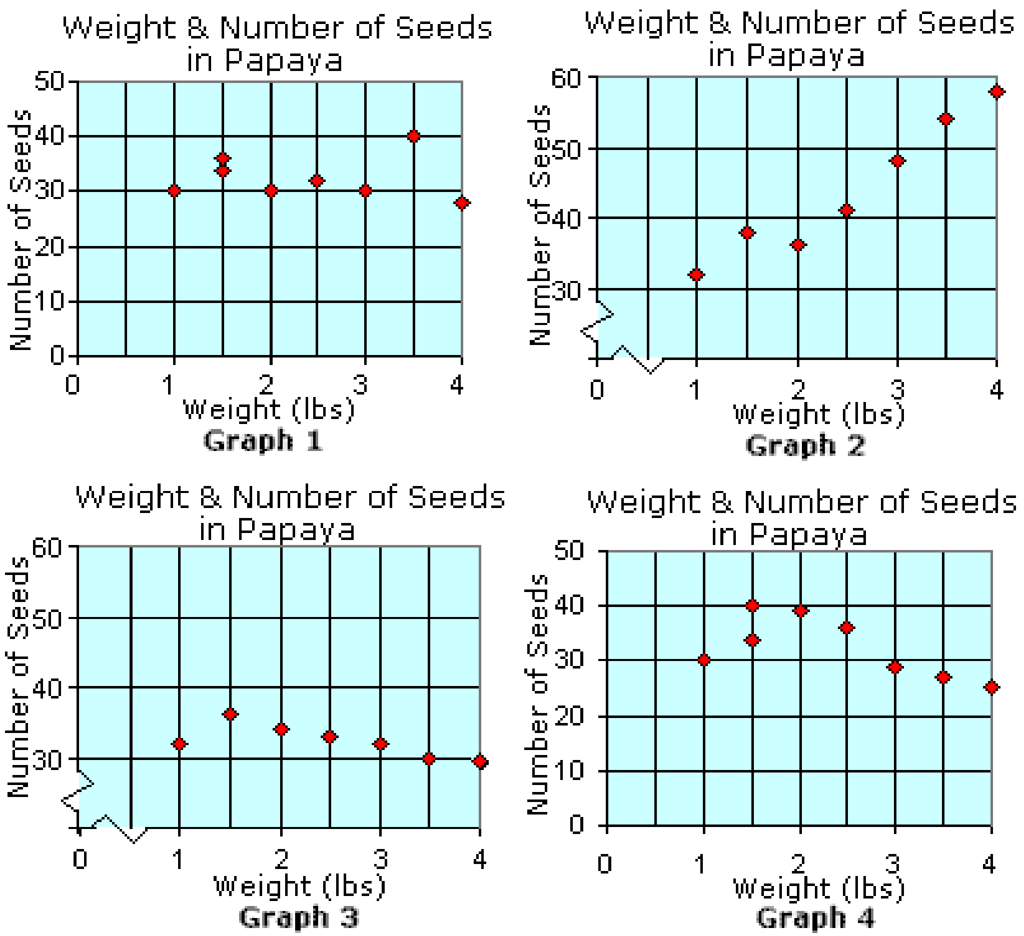
Used to identify the relation between variables, by plotting pairs of numerical data, with one variable on each axis

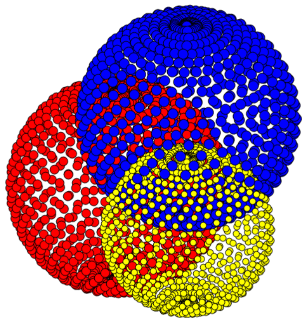
When to Use a Scatter Diagram

- When you have paired numerical data
- When your dependent variable may have multiple values for each value of your independent variable
- When trying to determine whether the two variables are related
  - When trying to identify potential root causes of problems
  - When determining whether two effects that appear to be related both occur with the same cause
  - When testing for autocorrelation before constructing a control chart



Example of Scatter Diagram





Cause and Effect Diagram

Also called Ishikawa or Fishbone Diagram

Used to structure the brain Storming Session – to sort ideas into useful categories

Many Possible Causes are identified for a stated problem and the effect on the problem are identified

When to Use a Cause and Effect Diagram

- **When identifying possible causes for a problem**
- **Especially when a team's thinking tends to fall into ruts**

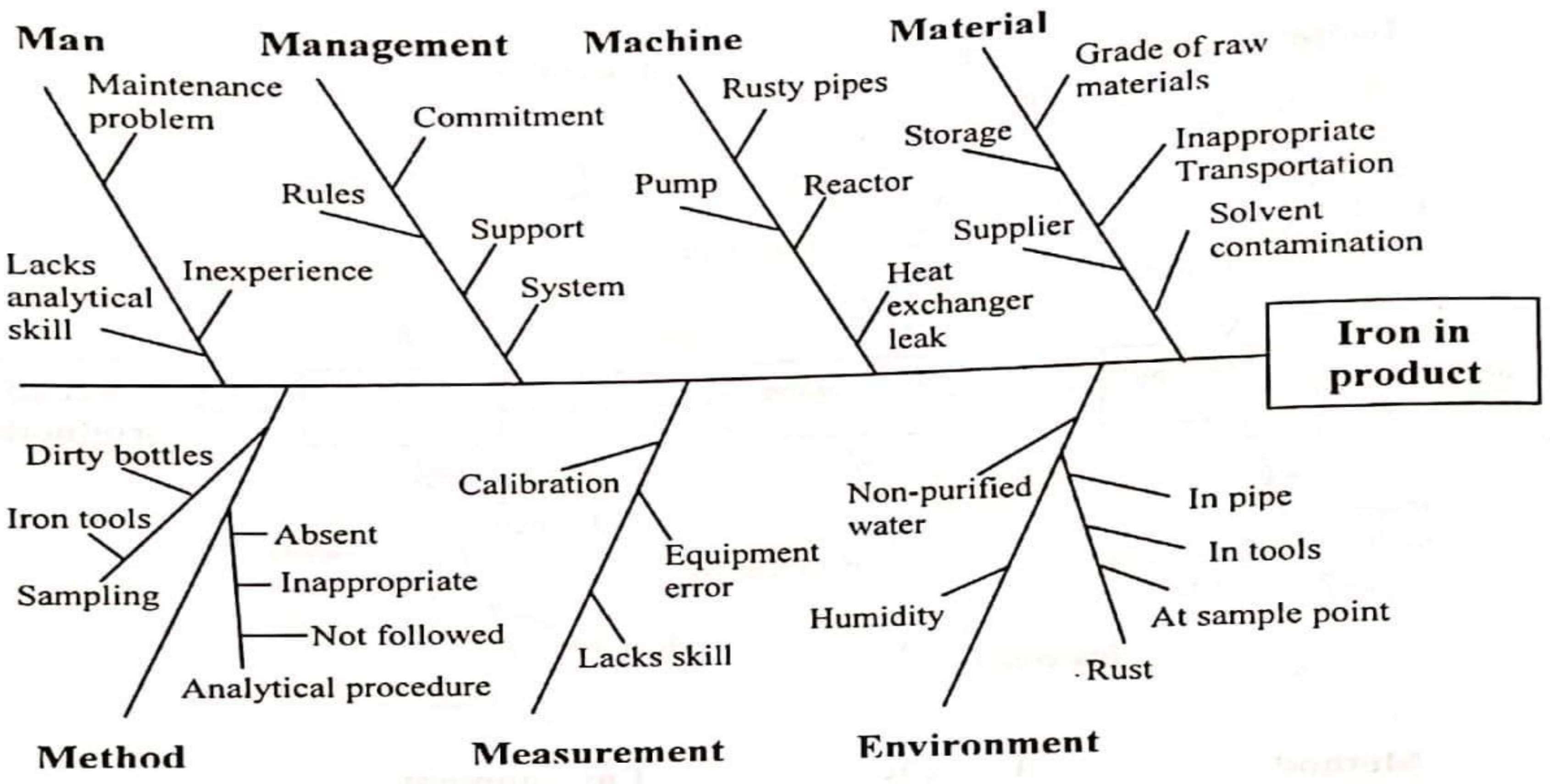
Types of CE Diagram

- **Cause Enumeration**
- **Process Analysis**

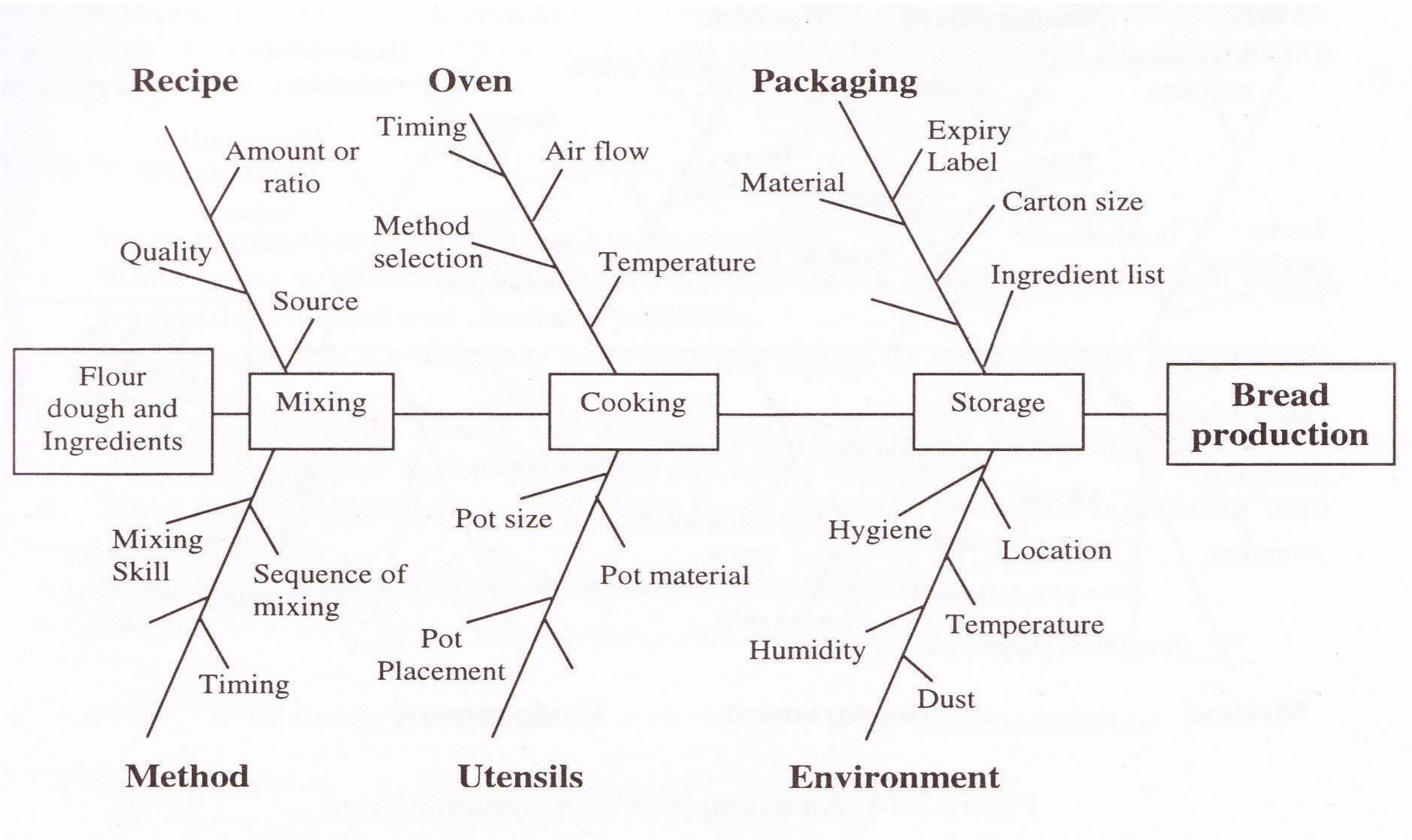
Cause Enumeration: identifies all possible causes from brainstorming sessions and then classifies into groups

Process analysis: follows the process step by step and causes are listed as per process step

Example of Cause Enumeration



Example of Process Analysis



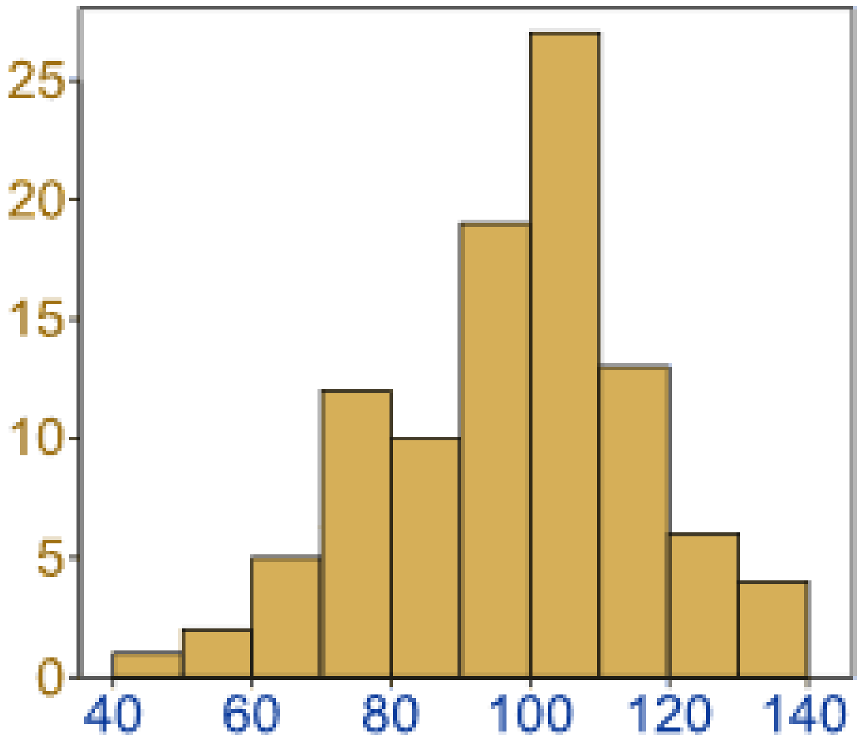
Histogram

Used to identify the frequency of occurrence of a variable in a set of data – looks very much like a bar chart

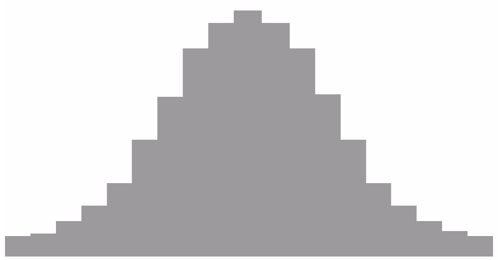
When to Use a Histogram

- When the data are numerical
- When you want to see the shape of the data's distribution, especially when determining whether the output of a process is distributed approximately normally
- When analyzing what the output from a supplier's process looks like
- When seeing whether a process change has occurred from one time period to another
- When determining whether the outputs of two or more processes are different
- When you wish to communicate the distribution of data quickly and easily to others

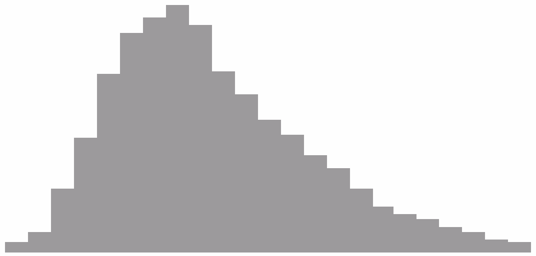
Example of Histogram



Typical shapes of Histogram

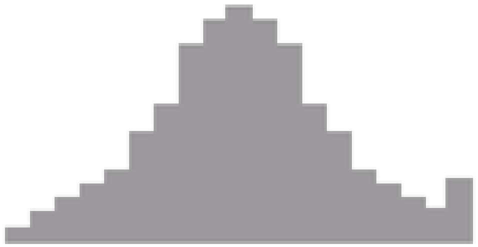


Normal distribution



Right-skewed distribution

Bimodal (double-peaked) distribution



Edge peak distribution

Comb distribution

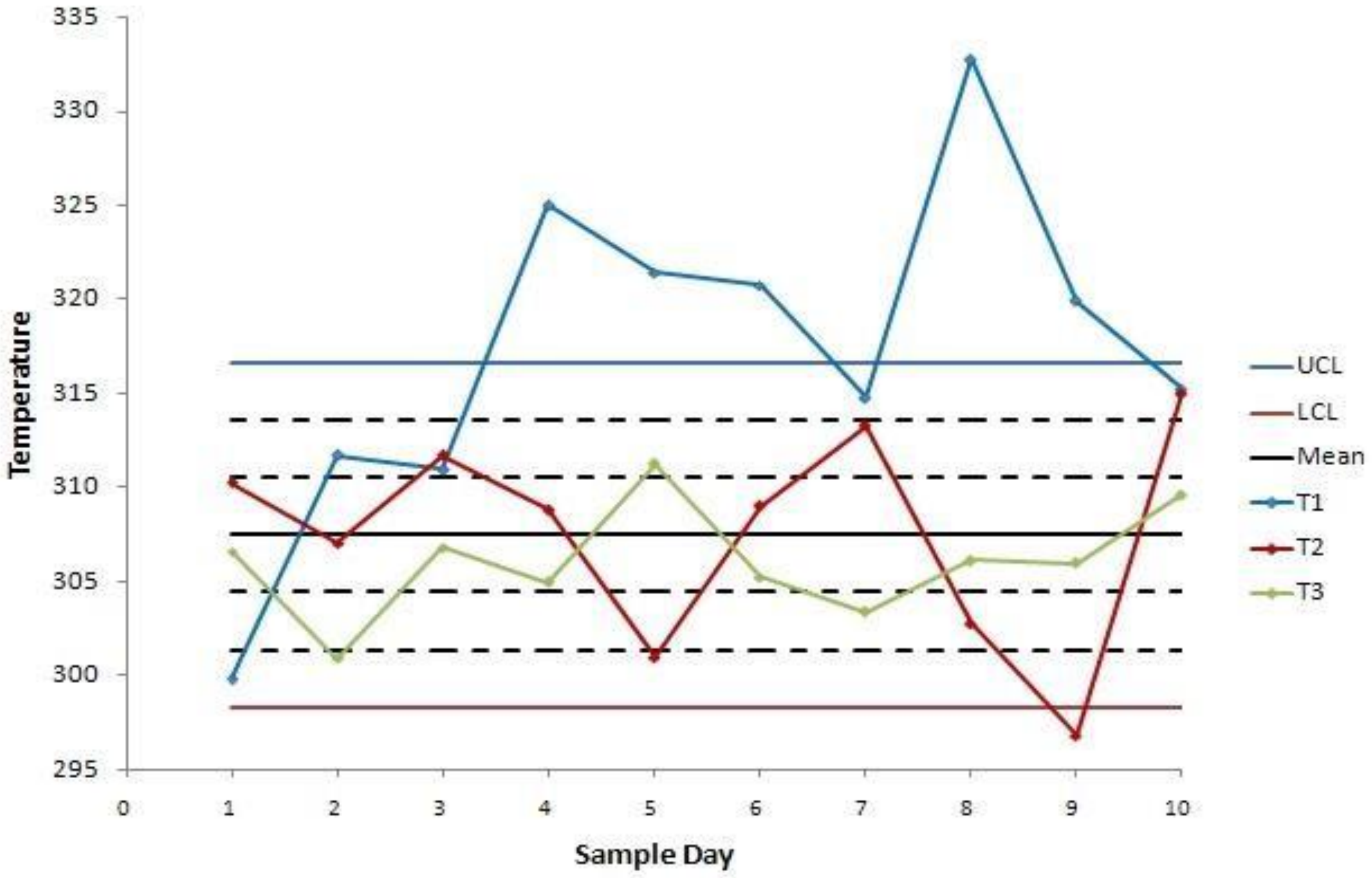
Control Charts

A graphical technique to study the changes to a process over time

When to Use a Control Chart

- When controlling ongoing processes by finding and correcting problems as they occur
- When predicting the expected range of outcomes from a process
- When determining whether a process is stable (in statistical control)
- When analyzing patterns of process variation from special causes (non-routine events) or common causes (built into the process)
- When determining whether your quality improvement project should aim to prevent specific problems or to make fundamental changes to the process

Control Charts



Control Charts for Variables

$\bar{X}$-R Chart

- 20-25 samples are taken usually each of which contains 4-6 observations
- If m samples are taken and m observations are made in each sample then the best estimator of process mean is the grand average.

$$\bar{\bar{x}} = \frac{\bar{x}_1 + \bar{x}_2 + \cdots + \bar{x}_m}{m}$$

- Where are the average of each sample
- If *x1, x2, . . . , xn* is the observations for sample size *n, then the range of the sample is the difference between the largest and* smallest observations; that is,

$$R = x_{\max} - x_{\min}$$

$$\bar{R} = \frac{R_1 + R_2 + \cdots + R_m}{m}$$

Control Charts for Variables

Control Limits for the $\bar{x}$ Chart
$\text{UCL} = \bar{\bar{x}} + A_2\bar{R}$
$\text{Center line} = \bar{\bar{x}}$
$\text{LCL} = \bar{\bar{x}} - A_2\bar{R}$

Control Limits for the R Chart
$\text{UCL} = D_4\bar{R}$
$\text{Center line} = \bar{R}$
$\text{LCL} = D_3\bar{R}$

Control Charts Factors

Subgroup	X bar chart		Sigma estimate	R chart		S chart	
	A_2	A_3	d_2	D_3	D_4	B_3	B_4
2	1.880	2.659	1.128	-	3.267	-	3.267
3	1.023	1.954	1.693	-	2.574	-	2.568
4	0.729	1.628	2.059	-	2.282	-	2.266
5	0.577	1.427	2.326	-	2.114	-	2.089
6	0.483	1.287	2.534	-	2.004	0.030	1.970
7	0.419	1.182	2.704	0.076	1.924	0.118	1.882
8	0.373	1.099	2.847	0.136	1.864	0.185	1.815
9	0.337	1.032	2.970	0.184	1.816	0.239	1.761
10	0.308	0.975	3.078	0.223	1.777	0.284	1.716
11	0.285	0.927	3.173	0.256	1.744	0.321	1.679
12	0.266	0.886	3.258	0.283	1.717	0.354	1.646
13	0.249	0.850	3.336	0.307	1.693	0.382	1.618
14	0.235	0.817	3.407	0.328	1.672	0.406	1.594
15	0.223	0.789	3.472	0.347	1.653	0.428	1.572
16	0.212	0.763	3.532	0.363	1.637	0.448	1.552
17	0.203	0.739	3.588	0.378	1.622	0.466	1.534
18	0.194	0.718	3.640	0.391	1.608	0.482	1.518
19	0.187	0.698	3.689	0.403	1.597	0.497	1.503
20	0.180	0.680	3.735	0.415	1.585	0.510	1.490
21	0.173	0.663	3.778	0.425	1.575	0.523	1.477
22	0.167	0.647	3.819	0.434	1.566	0.534	1.466
23	0.162	0.633	3.858	0.443	1.557	0.545	1.455
24	0.157	0.619	3.895	0.451	1.548	0.555	1.445
25	0.153	0.606	3.931	0.459	1.541	0.565	1.435

Control Chart Example

Table 11.1: $\overline{X}$ and R Values for steel pipe diameters (in centimeters).

Day	$\overline{X}$	R	Day	$\overline{X}$	R
1	10.724	0.040	12	10.730	0.026
2	10.730	0.016	13	10.735	0.028
3	10.718	0.040	14	10.726	0.041
4	10.728	0.014	15	10.724	0.025
5	10.730	0.027	16	10.720	0.017
6	10.720	0.020	17	10.727	0.035
7	10.720	0.038	18	10.720	0.037
8	10.711	0.026	19	10.726	0.030
9	10.713	0.027	20	10.724	0.012
10	10.718	0.008	21	10.718	0.030
11	10.717	0.039	22	10.722	0.012
			Total:	235.901	0.608

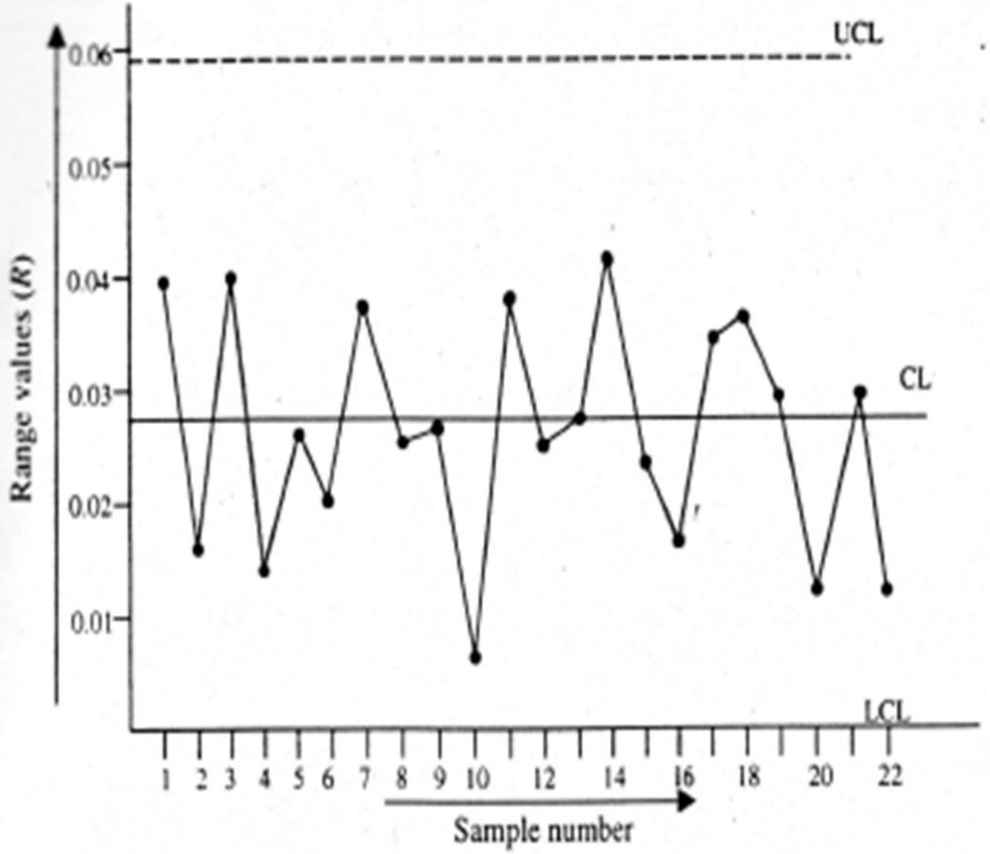


Figure 11.2: *R* chart for the example of steel pipe diameters.

It can be observed that none of the sample range values go out side of the limits. They are pretty random around the mean as well. Thus, this range chart can be considered as 'in-control' chart. As such, the $\overline{X}$ chart can now be developed.

While plotting this chart, the first step is to compute $\overline{X}$ using equation (2) –

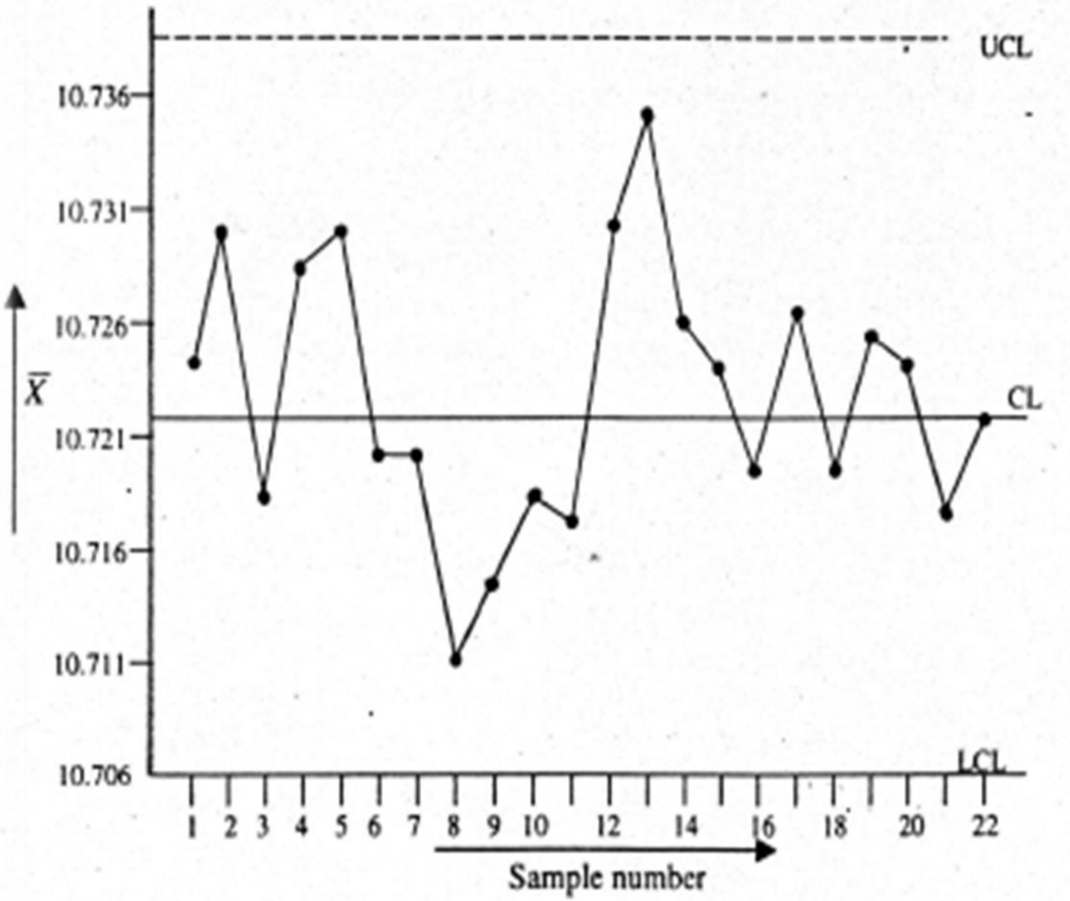


Figure 11.3: $\overline{X}$ control chart for the example of steel pipe diameter.

The $\overline{X}$ chart in Figure 11.3 reveals that the process is in-control on an aggregate basis, although certain points (sample number 7 to 14) indicate slight instability in the process. Nevertheless, the process is fairly random around the mean. It is, thus, concluded that the trial control limits can be adopted for future time periods.

Thank You!

Reference: Quality Control and Management(1st edition) By Ahsan Akhter Hasin